



Tractive Battery IPRO

Final Design Report

IIT Motorsports

Illinois Institute of Technology

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Contents

Summary	2
1 Synthetic Data Generation	3
1.1 Objective	3
1.2 Background	3
1.3 Methods	3
1.4 Calculations	3
1.5 Discussion	4
2 Cell Testing	5
2.1 Objective	5
2.2 Background	5
2.3 Methods	5
2.4 Calculations	5
2.5 Discussion	5
3 High Voltage Design	6
3.1 Objective	6
3.2 Background	6
3.3 Methods	6
3.4 Calculations	7
3.5 Discussion	7
4 Aerodynamic Simulation	7
4.1 Objective	7
4.2 Background	7
4.3 Methods	7
4.4 Calculations	8
4.5 Discussion	9
5 Thermal Simulation and Management	9
5.1 Objective	9
5.2 Background	9
5.3 Methods	9
5.4 Calculations	9
5.5 Discussion	10
6 PCB Design	10
6.1 Objective	10
6.2 Background	10
6.3 Methods	11
6.4 Calculations	11
6.5 Discussion	11
7 Fuse Design	11
7.1 Objective	11
7.2 Background	11
7.3 Methods	12
7.4 Manufacturing	12
7.5 Testing	12
7.6 Discussion	13
FSAE Rules Reference	14

Summary

This report documents the work completed by the IIT Motorsports IPRO course in support of the design and development of the 2027 IIT Motorsports FSAE electric racecar's Tractive Battery System. The IPRO was structured around eight sub-teams, each tasked with a specific aspect of the battery pack design process, with the collective goal of producing a set of validated design inputs and supporting analyses for use by the IIT Motorsports team in future development.

The Synthetic Data Generation team produced a physics-based simulation of the vehicle competing in a full 22-lap FSAE endurance event, generating time-resolved speed, current, voltage, and power profiles that served as primary inputs for several other sub-teams. The Cell Testing team physically discharged and characterized a sample of the 700 Molicel P45B cells in possession of IIT Motorsports, confirming that despite years of uncontrolled storage, the cells remain within an acceptable range of manufacturer specifications. Building on these results, the High Voltage Design team determined an optimal battery configuration of 84s-5p, producing a peak voltage of 296 V and a capacity of 6.7 kWh, and validated this configuration against the simulated endurance load using a custom Simulink model.

On the thermal side, the Thermal Simulation and Management team used Ansys and GT-AutoLion to characterize the heat generated by both individual cells and the full pack during the endurance event, and validated a cooling plate design capable of removing that heat via a liquid cooling loop. The Aerodynamic Simulation team complemented this work by evaluating five sidepod designs in Ansys Fluent, ultimately selecting a straight, longer sidepod with an undertray as the optimal geometry for directing airflow through the radiator while minimizing drag.

The PCB Design team assisted the IIT Motorsports Low Voltage team in the validation and layout of several PCB modules critical to the car's electrical architecture, including the Charging Module, Propulsion Control Module, and Shutdown Module. Finally, the Fuse Design team completed a full ground-up design, simulation, and physical testing process for a rules-compliant fusible link geometry for use in the 2027 battery pack busbars, establishing a repeatable process for future seasons.

Across all sub-teams, this IPRO produced a collection of validated analyses, physical test data, and design recommendations that directly advance the readiness of the 2027 IIT Motorsports battery pack. Areas identified for continued development include refinement of the endurance simulation model, further cell characterization testing, and completion of the remaining PCB modules.

1. Synthetic Data Generation

1.1 Objective

The objective of the Synthetic Data Generation team, which was comprised of Dina Liqa, Wiktor Ucher, Van Luan Khanh Nguyen, and Mongjae Ko, was to produce a simulated vehicle telemetry dataset representative of the IIT Motorsports FSAE electric racecar competing in a full 22-lap endurance event. Because the team lacked recent real-world telemetry from a completed vehicle, a physics-based computational approach was used to synthesize speed, current, voltage, and power profiles for use by downstream design teams.

1.2 Background

IIT Motorsports did not have recent on-car telemetry available to support the design of the 2027 electric vehicle. To address this gap, the team assembled vehicle specifications, motor parameters, and track geometry data to construct a digital vehicle profile. The endurance event was selected as the primary simulation target, as it represents the most demanding sustained operating condition for the Tractive Battery System and would allow other subteams to determine the voltage, current, and energy capacity requirements needed to complete the event.

1.3 Methods

The team's approach was structured as a two-phase workflow. In the first phase, vehicle and motor parameters were compiled and OptimumLap was selected as the primary lap simulation platform, as it is freely available tool widely used among FSAE teams. While the official track was being reconstructed, preliminary simulations were run using an existing Michigan track from OptimumLap's built-in library to begin validating the vehicle model.

The track reconstruction involved establishing a scale reference from the official 2023 event map (Figure 1), recreating the geometry in SolidWorks to extract segment lengths and corner radii, and inputting those dimensions into OptimumLap. In the second phase, the finalized track was used to generate a single-lap speed profile. A MATLAB model was then developed to extend the simulation across all 22 laps and apply the electrical conversion equations at each timestep.

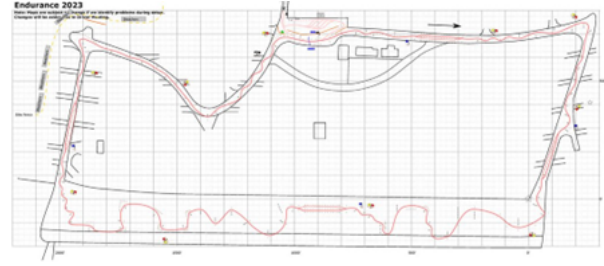


Figure 1: Official 2023 FSAE Michigan Endurance track map used as the reference for reconstruction.

The color-coded speed profile in Figure 2 illustrates vehicle speed around the reconstructed track, with red indicating higher speeds and green indicating braking and cornering zones.

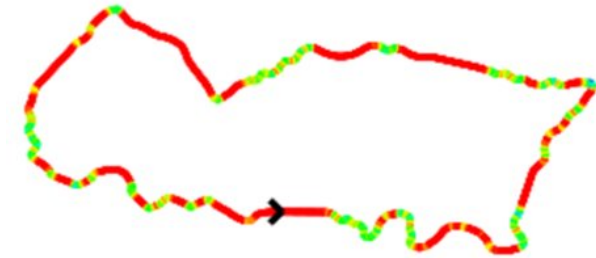


Figure 2: OptimumLap color-coded speed profile of the 2023 Michigan Endurance track.

1.4 Calculations

The vehicle, drivetrain, and motor parameters used as inputs to both OptimumLap and the MATLAB model are summarized in Table 1.

Table 1: Vehicle and system parameters.

Parameter	Symbol	Value
Vehicle mass (with driver)	m	294.8 kg
Drag coefficient	C_d	1.44
Frontal area	A	1.054 m ²
Air density	ρ	1.2 kg/m ³
Rolling resistance coeff.	C_{rr}	0.015
Tire radius	r	0.203 m
Gear ratio	GR	4:1
Drivetrain efficiency	η	0.92
Torque constant (EMRAX 188)	K_t	0.33 Nm/A
Max pack voltage	V_{max}	270 V
Max motor speed	RPM_{max}	7,000 RPM
Peak motor torque	T_{max}	100 Nm
Continuous motor torque	T_{cont}	50 Nm
Peak motor current	I_{peak}	300 A
Continuous motor current	I_{cont}	150 A

Current demand at each timestep was derived from the vehicle speed profile using:

$$I = \frac{(ma + \frac{1}{2}\rho C_d A v^2 + C_{rr} m g) \cdot r}{GR \cdot \eta \cdot K_t} \quad (1)$$

Terminal voltage was modeled as proportional to motor shaft speed via a back-EMF relationship:

$$V = \frac{V_{max} \cdot \text{speed} \cdot 60 \cdot GR}{3.6 \cdot 2\pi r \cdot RPM_{max}} \quad (2)$$

Instantaneous battery power was computed as $P = V \times I$, and total energy was obtained by integrating power over the full event duration.

1.5 Discussion

The simulation produced time-resolved current, voltage, and power profiles across the full 22-lap event. Key results are summarized in Table 2.

Table 2: Simulation output summary.

Metric	Value
Lap time	117.7 s
Peak current	256.4 A
Peak voltage	270.0 V
Minimum voltage	54.9 V
Total energy required (22 laps)	21.720 kWh
Usable battery energy (assumed)	21.578 kWh
Energy deficit (ΔE)	0.142 kWh
Average continuous power	≈ 30.2 kW

The current profile showed repeated peaks approaching 256 A during acceleration zones and corner exits, as shown in Figure 3. Voltage tracked motor speed closely, reaching 270 V at peak speed and dropping to approximately 55 V at the slowest corners (Figure 4).

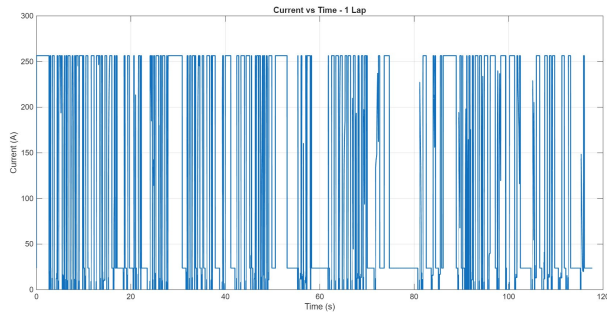


Figure 3: Instantaneous motor current over one lap. Peaks near 256 A correspond to acceleration and corner exit zones.

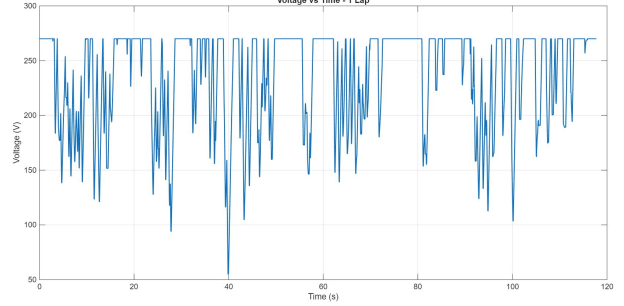


Figure 4: Terminal voltage over one lap, tracking motor speed.

The simulation indicated a marginal energy deficit of 0.142 kWh over 22 laps. While this technically suggests the vehicle cannot complete the endurance event on a single charge, the team considers this result to be within the uncertainty of the model. Several potential sources of error may account for this gap. The vehicle parameters used throughout are preliminary estimates based on design targets and literature values, and may differ from the final as-built configuration. Additionally, the back-EMF voltage model is a simplification of real motor behavior, and inaccuracies in this relationship would compound across all 22 laps in the multi-lap simulation. The track geometry, while reconstructed from the official event map, is also an approximation and may not perfectly reflect the true lap distance and corner profile. Given these sources of uncertainty, the 0.142 kWh deficit cannot be taken as a definitive conclusion that the vehicle would fail to complete the event. As this work continues to be developed, refinement of the vehicle model by incorporating more accurate as-built parameters would allow for a more confident determination of whether the battery is sufficient to complete a full endurance event.

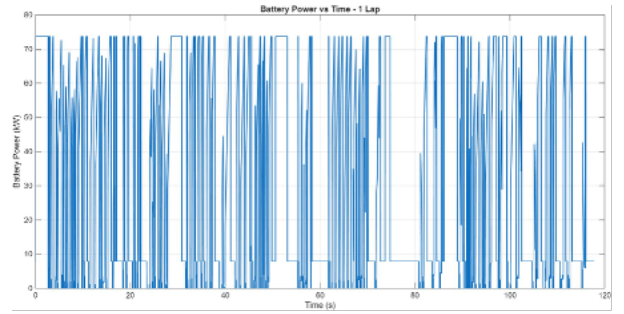


Figure 5: Battery power demand over the full 22-lap endurance event. Average power ≈ 30.2 kW with peaks near 75 kW.

2. Cell Testing

2.1 Objective

The cell testing team, which was comprised of Bartlomiej Potok and Antonio Flores, was tasked with the collection and verification of the Molicel P45B cell characteristics based on the batteries stored in the IIT Motorsports garage.

2.2 Background

The IIT Motorsports team came into ownership of approximately 700 Molicel P45B 21700 lithium battery cells in the fall semester of 2024. These cells ended up going unused through both the 2025 and 2026 season and sat in their original packaging on shelves in the IIT Motorsports garage. While the garage is heated in the winter, there can be considerable temperature swings, with temps in the summers of over 90 degrees F. These conditions can be known to cause adverse effects on cell lifetime, capacity, and internal resistance. Along with the unknown degradation due to temperature, the cells were left as they were on arrival, with no measures taken to ensure that the voltage level was not too high for long term storage.

Coming to today, there is now a desire to utilize these cells in the next battery pack for the 2027 IIT Motorsports car. Given the unknown state of the cells, it was decided to record the maximum capacity, open circuit voltage (OCV), and internal resistance of a random assortment of cells from different boxes to understand how much deviation from the characteristics on the data sheet has occurred.

2.3 Methods

The testing of the cells was accomplished primarily via the use of a DWEII DW-124712 18650 battery tester [1]. Due to the fact that the P45B is a 21700 cell, an additional 21700 battery holder was employed [2] to connect the cell to the tester. A supplementary Fluke multimeter was used to double-check voltage and resistance measurements. The DW-124712 was first used to charge the battery to be as close to 4.2 Volts as possible. This voltage was then recorded as the 100% State-of-Charge (SOC). The DW-124712 was then set to discharge the battery at an initial 1 Amp via an internal resistor of about 3.94 Ohms. Measurements of the OCV with the discharge test paused and the voltage with the load connected were taken at intervals of approximately 10 minutes using both the voltage measuring features of the DW-124712 and the Fluke multimeter. This process continued until the battery reached a voltage of 2.5V. Tests were performed on a total of 14

cells across all 6 boxes. Each cell trial took around 5 hours to complete.

2.4 Calculations

Using the collected data (OCV, V_{load} , load resistance) the SOC and internal resistance were calculated using the following equations.

$$SOC(t) = SOC_0 - \left(\frac{I(t)}{C_{nom}} \right) \times t \quad (3)$$

$$R_i = \left(\frac{V_{oc} - V_{load}}{I} \right) \times RL \quad (4)$$

2.5 Discussion

After the completion of all physical cell tests, the data was aggregated on an Excel sheet and averaged to find the new nominal maximum characteristics for the cells in possession of the IIT Motorsports team. One such trial is seen in Table 3.

Table 3: Molicel P45B Cell 1a discharge test data — Box 1.

Time	C (mAh)	SOC	V_{oc}	V_{load}	R_i (Ω)	I (A)
0:00:00	4307	100.00%	4.17	3.93	0.01566	0.99
0:10:12	4140	96.12%	4.10	3.88	0.01454	0.98
0:20:06	3979	92.38%	4.08	3.87	0.01391	0.97
0:30:04	3818	88.65%	4.06	3.86	0.01329	0.97
0:40:04	3659	84.95%	4.02	3.82	0.01342	0.96
0:50:08	3498	81.22%	4.00	3.79	0.01421	0.96
1:00:07	3343	77.62%	3.96	3.73	0.01581	0.94
1:10:08	3186	73.97%	3.91	3.70	0.01455	0.93
1:20:06	3033	70.42%	3.87	3.66	0.01471	0.92
1:30:05	2880	66.87%	3.84	3.64	0.01409	0.92
1:40:06	2728	63.34%	3.82	3.61	0.01492	0.91
1:50:29	2572	59.72%	3.79	3.58	0.01504	0.90
2:00:11	2426	56.33%	3.76	3.57	0.01365	0.90
2:10:05	2279	52.91%	3.73	3.54	0.01376	0.89
2:20:07	2132	49.50%	3.70	3.48	0.01621	0.88
2:30:07	1988	46.16%	3.67	3.45	0.01635	0.86
2:40:16	1843	42.79%	3.63	3.42	0.01574	0.86
2:50:14	1699	39.45%	3.60	3.40	0.01508	0.86
3:00:11	1558	36.17%	3.57	3.37	0.01522	0.85
3:10:30	1414	32.83%	3.53	3.34	0.01459	0.84
3:20:06	1280	29.72%	3.50	3.31	0.01472	0.83
3:38:04	1033	23.98%	3.45	3.26	0.01494	0.82
3:40:10	1003	23.29%	3.44	3.25	0.01499	0.82
3:50:07	869	20.18%	3.39	3.21	0.01438	0.81
4:00:12	735	17.07%	3.33	3.16	0.01379	0.79
4:10:07	606	14.07%	3.27	3.08	0.01582	0.78
4:20:30	474	11.01%	3.19	3.01	0.01533	0.76
4:30:17	352	8.17%	3.12	2.94	0.01570	0.74
4:40:07	234	5.43%	3.04	2.84	0.01806	0.71
4:50:09	117	2.72%	2.90	2.72	0.01697	0.68
5:00:10	6	0.14%	2.74	2.54	0.02019	0.64
5:00:49	0	0.00%	2.71	2.50	0.02154	0.63

In comparing the results with the datasheet from Molicel [3], while the battery exhibits a voltage and capacity at the lower end of the range — at about 4.17V and 4300 mAh respectively compared to the expected 4.2V and 4500 mAh — it is still well within

the expected range. In addition, the measured internal resistance across the discharge tests also was in-line with expected values despite the measurement equipment being somewhat incapable of taking fine readings at the scale of tens of milliohms; this is reflected in the jumps between datapoints seen in Table 3.

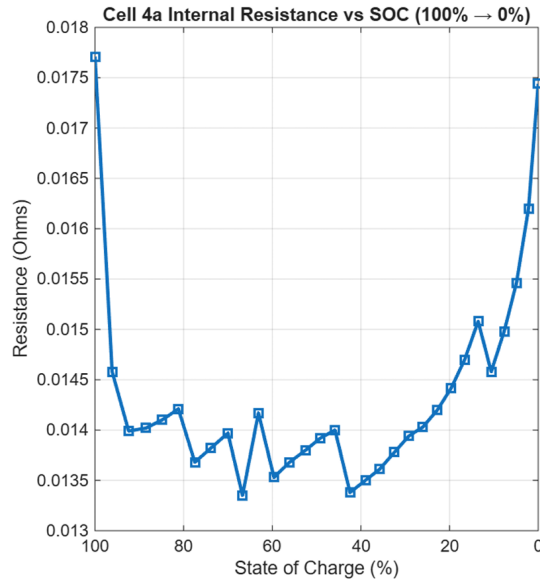


Figure 6: Molicel P45B calculated internal resistance vs. SOC for Cell 4a.

In total, while this test was by no means an exhaustive dive into the characteristics of the cells on hand, it was able to ensure that the cells intended for use in the 2027 car are still within an acceptable range of the manufacturer specifications. Future, more rigorous, testing could include the variation of temperature during discharge, as well as the cycling of multiple charge-discharge cycles on the same cell. The data collected here directly influenced the choices made by the High Voltage Design team later in this report and will be used by future members of the IIT Motorsports team in any use of the stockpiled batteries.

3. High Voltage Design

3.1 Objective

The objective of the High Voltage Design team, which was comprised of Liliia Shanina, Cruz Herrera, and Franco Avalos, was to calculate an appropriate battery configuration based on 2027 car parameters, simulated load requirements, and physical space constraints.

3.2 Background

An optimal battery pack design for a Formula Student racecar must meet numerous operational and rules criteria. Besides rules constraints, some of the most important design requirements come from the choice of cell, motor, and inverter. A motor will determine what operational voltage and current is required for the pack; the inverter will determine what peak voltage is required to stay within the operating ranges of the device, and the cell determines the operational limits of a given configuration. In the context of this IPRO, all three of these factors have already been decided, as the IIT Motorsports team is already in possession of the Emrax 188 motor, Cascadia PM100DX inverter, and cell choice of the Molicel P45B. Given this hardware configuration, the HV team must then come up with a configuration that meets the capacity requirements of the endurance test, as well as meets all other rule constraints.

3.3 Methods

The team first considered all parameters given by the hardware choices. P45B cells have a max voltage of 4.2 V, 45 A max discharge, and 4.5 Ah capacity. The inverters have a hard voltage limit of 420 V. Each 188 motor has a peak current demand of 300 A. Then, the team considered the competition limit of 80 kW maximum energy output at any given moment. The team then created a script based on the parameters above combined with equations in the calculations section to come up with a viable battery configuration. This configuration, which was settled on, was 84s-5p split into 6 14s-5p segments. As seen in Table 4, this configuration gives a peak voltage of 296 V and output current of 227 A. The pack also has a capacity of 6.7 kWh.

Table 4: Battery configuration parameters and comparison.

	Last Year (108S-4P)	Current (84S-5P)
Peak Voltage	453 V	296 V
Current	180 A	227 A (+26.1%)
Cells	432	420 (-12)
Volume (cells)	2698 in ³	2623 in ³

After the initial pack configuration design given the flat hardware and rules constraints, the team then produced a Simulink model which could accept the power demand created by the Data Generation team. This Simulink model would then calculate the resulting SOC, voltage and current outputs, and power as a function of time. Concurrently, the team also created a preliminary SolidWorks model of the pack in con-

junction with members of the Thermal Management team, as seen in Figure 7.

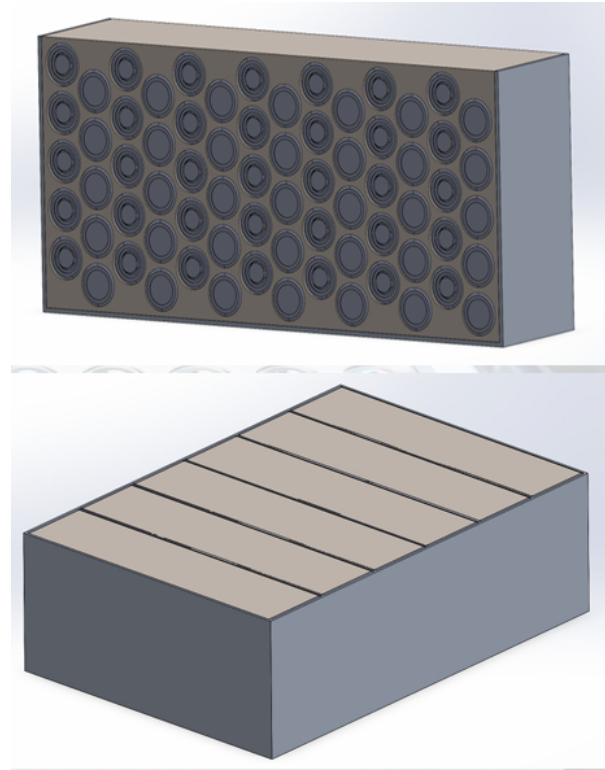


Figure 7: Preliminary battery 3D assembly based on 84s-5p configuration.

3.4 Calculations

The MATLAB script used for the battery configuration calculations and Simulink model used for the realistic power demand simulation can be found in the OneDrive. Figure 8 shows the block diagram of the Simulink model.

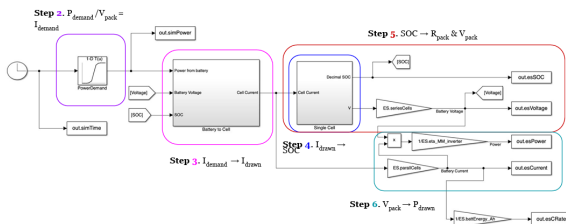


Figure 8: High-level block diagram of pack Simulink model.

3.5 Discussion

This team was successful in creating a battery pack configuration compatible with all hardware and rules requirements given by IIT Motorsports. Where this

design did fail to meet expectations was the simulated load conditions of the endurance track given by the Synthetic Data team. In the simulated run, the battery was drained of energy before completing all 22 laps. While this could appear to be solely an issue of the capacity of the pack, it very much could be that the load conditions provided were too demanding for any rules compliant battery pack to fill under the hardware constraints. Further investigation should be undertaken to determine whether a change in configuration can meet the current load conditions; otherwise, the load conditions and lap times may have to be adjusted down to meet the realistic output ability of the battery pack.

4. Aerodynamic Simulation

4.1 Objective

The objective of the Aerodynamic Simulation team, which was comprised of Alexander Rodriguez, Admir Pali, John Munoz, and Anna Elena Sofia Sebastio, was to find the optimal location and size of a radiator to remove heat from the cooling loop.

4.2 Background

Generally, the purpose of an aerodynamic package on a Formula Student vehicle is to provide downforce to increase cornering speeds. However, devices normally used for aerodynamic effects can sometimes be adapted to serve other purposes as well. In the past, IITM has done this by having the tractive battery system enclosed in the side pods. For this design, the intent was to use the side pod to concentrate air towards the radiator/heat exchanger in an effort to help direct the turbulent flow regime to be used for cooling.

4.3 Methods

To determine what type of side pod would produce the best mass flow rate through the radiators, 5 different side pods were designed and simulated using Ansys Fluent: (1) a straight, short sidepod,

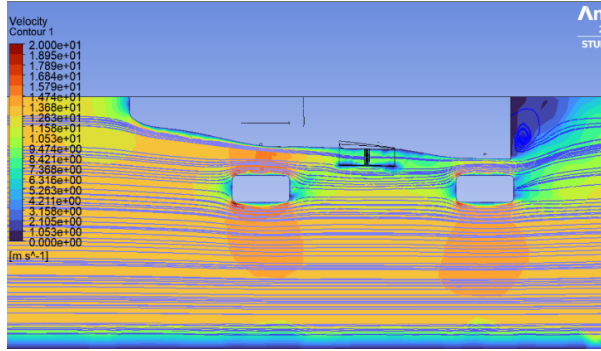


Figure 9: Top View of Design 1 Velocity Contour with Streamlines.

(2) a short sidepod positioned further downstream,

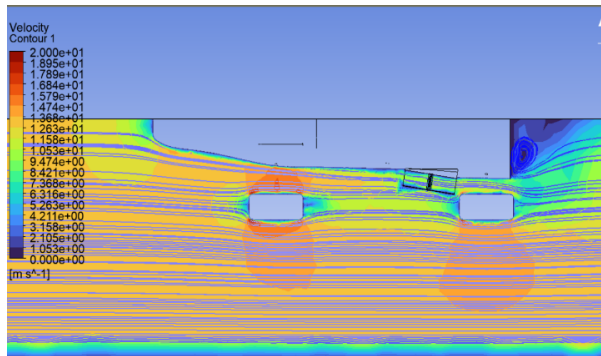


Figure 10: Top View of Design 2 Velocity Contour with Streamlines.

(3) a convergent inlet sidepod,

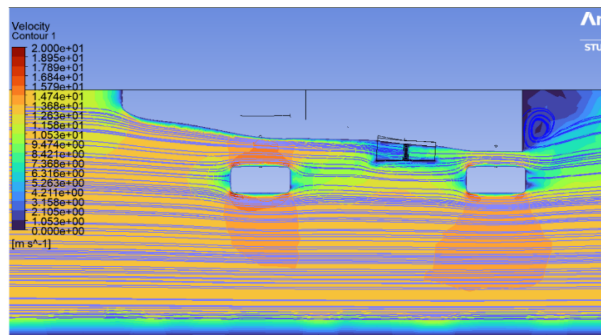


Figure 11: Top View of Design 3 Velocity Contour with Streamlines.

(4) a divergent inlet,

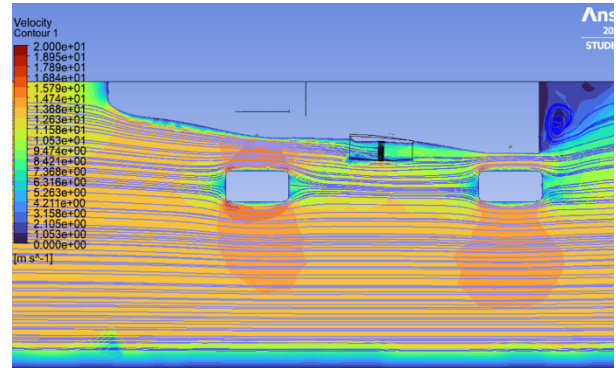


Figure 12: Top View of Design 4 Velocity Contour with Streamlines.

and (5) a straight, longer sidepod with an undertray.

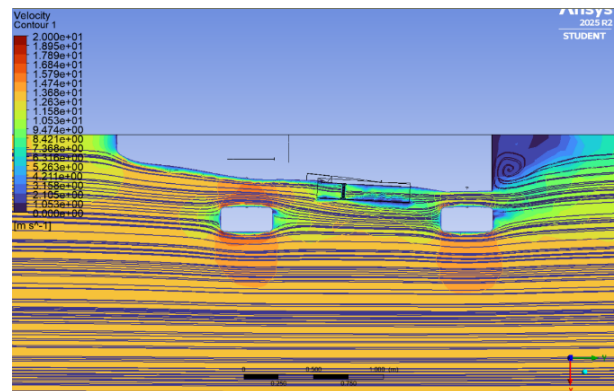


Figure 13: Top View of Design 5 Velocity Contour with Streamlines.

4.4 Calculations

Using the simulation and downforce data, a decision table was constructed (Table 5) to determine the optimal sidepod design. The table includes considerations for mass flow rate, change in pressure, downforce, and drag.

Table 5: Sidepod Decision Matrix.

Metric	Weight	Normalization	Design 1	Design 2	Design 3	Design 4	Design 5
$\dot{m}$	0.35	$x/0.325$	0.3420	0.3797	0.2280	0.3603	0.4399
Δp	0.25	$x/-40.8$	0.2451	0.2696	0.1838	0.2451	0.3064
Downforce	0.18	$x/-1.532$	0.2088	0.2145	0.2691	0.0965	0.1111
D	0.22	$11.832/x$	0.2224	0.2183	0.2034	0.2229	0.2355
<i>Total Weighted Score</i>			1.018	1.082	0.884	0.925	1.093

4.5 Discussion

The decision matrix determined that design 5 (the straight, longer sidepod with an undertray) would produce the optimal performance. In the individual categories, this design produced the highest mass flow rate (0.409 kg/s, STD = 0.0715 kg/s), second highest pressure differential (−50 Pa, STD = 7.0143 Pa), fourth highest downforce (−0.9453 N, STD = 0.6557 N), and the least drag (11.054 N, STD = 0.6390 N). As expected, the convergent inlet design exhibited the best aerodynamic performance (Lift/Drag = −0.179); however, this design produced the lowest values for mass flow rate and pressure differential, which are the main determinants of cooling performance. Design 5 had the highest values for both mass flow rate and pressure differential, and produced the least drag. Given that this sidepod is not being designed as an aerodynamic device (T.7.1.1), design 5 was chosen because it produced the best cooling values while keeping the drag added to a minimum.

5. Thermal Simulation and Management

5.1 Objective

The objective of the thermal management team, which was comprised of Jack Lin, Axel Barrera, and Yahir Vasquez Vallejo, was to design and validate a system to transfer heat from the battery pack to a cooling loop.

5.2 Background

In previous years, IIT Motorsports has used a Phase Change Composite (PCC) material to effectively remove heat from the battery. This material was no longer able to be used, causing the team to switch to a PVC filler for the batteries. Because PVC has a thermal conductivity of 0.15 W/m·K, it could not remove heat from the battery pack efficiently.

5.3 Methods

Thermal simulations were performed using Ansys to determine the amount of heat produced by a single cell over the course of the endurance event. A ther-

mal simulation was also performed simulating the overall heat produced by the battery pack. Additionally, GT-AutoLion was used to verify the thermal values based on the power draw from the battery during the endurance event.

To determine if a cooling plate design would be effective, two Ansys transient thermal simulations were done of the cooling plate: one with static internal heat values to measure the temperature of the cooling plate, and one with flowing water through the pipes in the cooling plate to measure the temperature of the water.

5.4 Calculations

Figure 14 shows the results of the transient thermal simulation of the cooling plate with the applied heat values from the thermal simulation team.

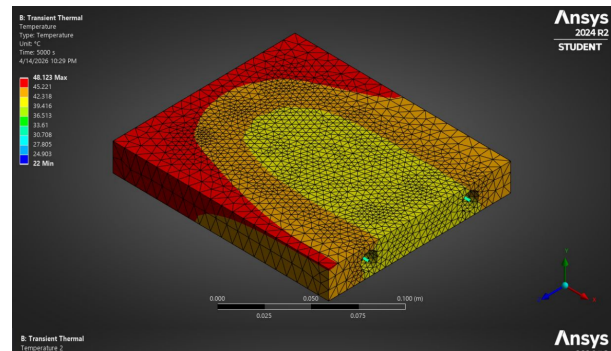


Figure 14: Transient Thermal Simulation of Cooling Plate with Applied Heat from Battery Pack.

Figure 15 shows the transient thermal simulation which used flowing water to measure the temperature as it flowed through the cooling plate.

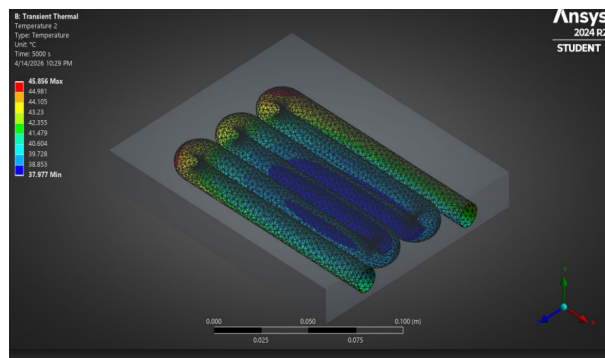


Figure 15: Transient Thermal Simulation of Flowing Water through the Cooling Plate.

Figure 16 shows the transient thermal simulation of a single cell using data from the synthetic data generation portion. Figure 17 shows the thermal simulation of the full pack using all of the cells and a container.

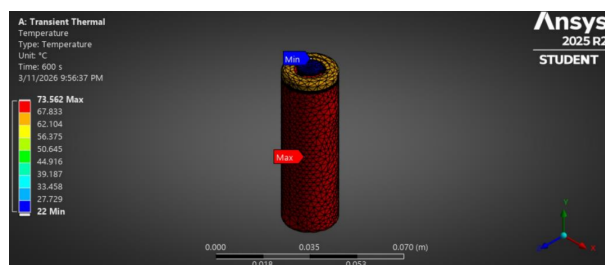


Figure 16: Single Cell Thermal Simulation.

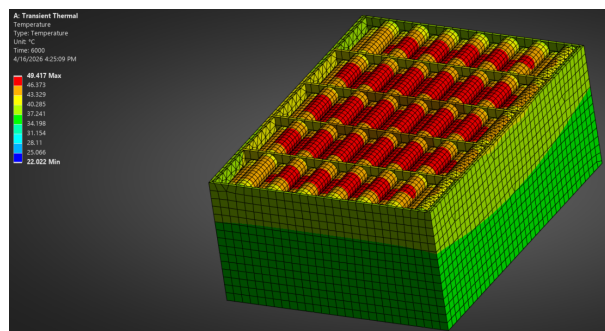


Figure 17: Full Pack Thermal Simulation.

5.5 Discussion

The results of the simulation indicate that the cooling plate and thermal setup will be able to effectively remove heat from the battery given that the individual modules are able to effectively remove heat from the battery cells and transfer it to the cooling plate. There was less confidence in the simulation

with flowing water because we would expect to see the temperature of the water rise continuously as it moves through the plate (which has a constant heat applied to the entire top face). However, the transient cooling plate simulation does show what we would expect to see, the areas closest to the water get cooled very effectively with a temperature gradient going up as you get closer to the outside of the cooling plate. The Full pack thermal simulation does have some limitations in its applicability because it used air between the cells, which will not be the case in the actual battery pack. However, because static air has a relatively low thermal conductivity, the simulation data shows that with a better thermal conductor, the heat would be able to be effectively transferred to the cooling plate.

6. PCB Design

6.1 Objective

The objective of the PCB Design team, which was comprised of Bhavik Singh, Karla Leonte, Arianna Perez, Mahmoud Hayek, Emiliano Mora, and Nathan Trujillo, was to assist the IIT Motorsports Low Voltage team in their development of custom Printed Circuit Boards (PCBs) for the 2026 season.

6.2 Background

The IIT Motorsports low voltage control architecture relies on several PCB modules dispersed around the race car and supporting equipment, such as the charger. These modules consist of the Shutdown Module (SM), Propulsion Control Module (PCM), Dashboard Module (DM), Charging Module (CM), Battery Monitoring System (BMS), Accumulator Monitoring System (AMS), High Voltage Board (HVB), and Power Distribution Module (PDM). The IIT Motorsports team during the second half of the 2026 season was significantly behind in the development of the PCBs required above in the low and high voltage control of the car. Due to a lack of manpower, it was decided to utilize the student resources of the IPRO to aid in the development of the PCBs.

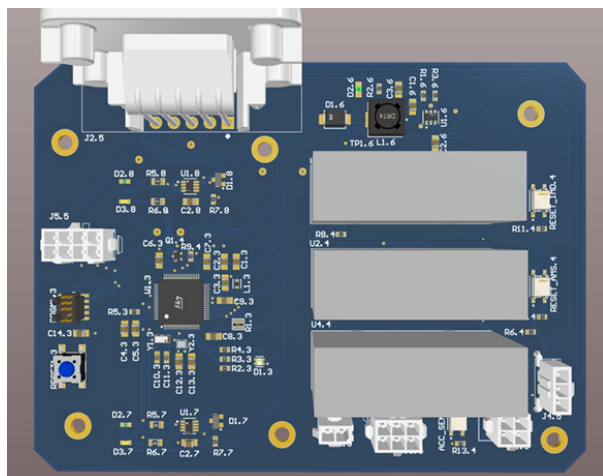


Figure 18: Render of complete CM PCB module.

6.3 Methods

The IIT Motorsports team uses Altium Designer as its preferred ECAD software. Most of the members of this team had not used the software before, and this required a significant introduction to PCB design. To bridge this gap, a video tutorial was provided [5]. Once completed, the members of this team were tasked with validating and correcting any issues on the CM, PCM, and SM PCBs, all of which were already in a state of completion from work done by other members of IIT Motorsports. Errors in routing and schematics were discovered using the Design Rules Checker (DRC) based on a list of constraints provided by the IIT Motorsports team. DRC errors include but are not limited to: trace spacing, trace intersections, acid traps, net antennas, unrouted nets, vias in pads, lack of annular rings, component spacing, and pour islands. For the sake of brevity, the intricacies of each error will not be discussed here, but in-depth guides and other provided resources produced by Bhavik Singh can be found in both the IIT Motorsports team OneDrive and the team Discord. Aside from error correction, some team members were able to lay out and route sections of the AMS, which had no such layout before the beginning of the IPRO. The process of laying out a PCB is directly related to a multitude of variables such as current requirements, datasheet recommendations, Electromagnetic Interference (EMI) concerns, physical space constraints, and many others. Team members also used the hierarchical block diagram to organize components in physical areas based on which circuit they were a part of.

6.4 Calculations

No quantitative calculations were performed as part of this effort; the work consisted primarily of design validation and layout within Altium Designer.

6.5 Discussion

The prospect of learning, let alone mastering ECAD and PCB design in the span of a 3-credit hour course is more than ambitious. Despite this, the PCB Design team was able to make a concerted effort in learning a skill that was new to a majority of members. Going further, the team was able to provide much needed support to the spread-thin IIT Motorsports Low Voltage team in their validation and layout of multiple modules.

7. Fuse Design

7.1 Objective

The objective of the fuse design team, which was comprised of Evan Pollack and Eman Sator, was the design, manufacture, and testing of a rules compliant fusible link design process for the 2027 battery pack.

7.2 Background

For the past few years, the battery pack design used by IIT Motorsports has been comprised of individual cylindrical cells connected using spot welded busbar plates like the design seen in Figure 19.

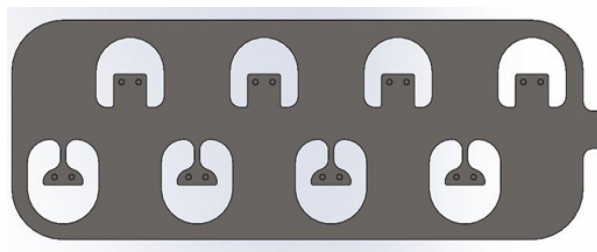


Figure 19: Single busbar plate for 4P configuration.

As required by (EV.6.6.3), each individual cell has overcurrent protection via a fusible link designed into the geometry of the busbar. The geometry of the link itself must be designed to adequately protect the pack in the event of an internal short in a cell, or a thermal runaway event that is not caught by the BMS. The geometry of the fusible link is dependent on the battery configuration, cell type, motor load, and the material of the busbar plate itself. In the past few seasons, the IIT Motorsports team has not had busbar plates with fusible links that meet competition rules specifications. It is for that reason that this team has been tasked with building a ground-up process for the design and verification of a fusible link

geometry. The parameters of the fusible links created for the purposes of the IPRO were based on the 4P configuration and the Molicel P45B cell of the 2026 IIT Motorsports car due to the 2027 configuration not being available at the beginning of this IPRO. In addition, the Formula Student Michigan competition requires test data for custom designed fusible links based on the information provided in [4].

7.3 Methods

Based on the 2026 battery configuration, the minimum failure mode current of the fusible link was given at around 80A. A curve of experimental data based on nickel silver was also provided as a baseline as seen in the blue line of Figure 20. The team then utilized a combination of SolidWorks and ANSYS Electric Transient Thermal to create and simulate fuse geometry. It was decided that nickel plated steel would be used for all the designs due to its availability, cost, solderability, and wide use as a busbar plate material in battery pack designs. After many rounds of simulation, a geometry of 0.4572 mm $\times$ 2.2 mm was settled upon, giving a theoretical failure curve as seen in the green line of Figure 20.

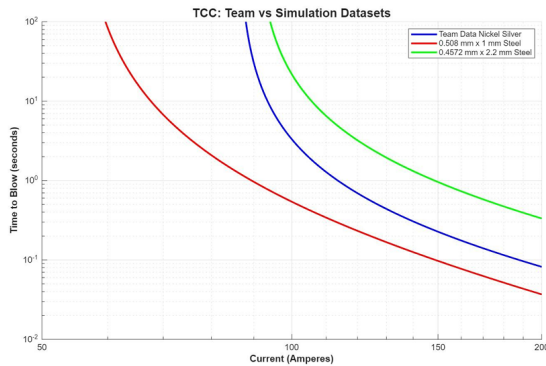


Figure 20: Simulation results of fuse design.

7.4 Manufacturing

Once the fuse geometry was chosen, the team then moved on to manufacturing samples for physical testing. Due to time and availability constraints, the team was unable to source nickel plated steel for this test and instead had to rely on unplated steel plating for the fusible link test pieces. While the nickel-plating may have some effect on the failure time of the fuses, it mainly serves as a means of soldering voltage taps to the outside of the busbar. Outside of the IPRO, members of the IITM team investigated electroplating nickel to better simulate the nickel-plated steel busbars intended for battery pack use. An unconventional electrolyte of Mon-

ster Zero Ultra™ was utilized successfully in tests of nickel-plating pennies. Once the 0.4572 mm steel plate arrived, around 20 fusible link test pieces were cut out from the sheet using a table router.

7.5 Testing

The testing of the fusible links was completed as per the Guidelines for the testing of team designed fusible links [4], in which 4 distinct failure timeframes must be met as seen in Figure 6.

Table 6: Fusible link testing failure times [4].

Blow Time	
Current Level 1	30 sec $< t <$ 300 sec
Current Level 2	5 sec $< t <$ 15 sec
Current Level 3	0.2 sec $< t <$ 1 sec
Current Level 4	0.05 sec $< t <$ 0.15 sec

As per the design, current level 1 was to be around 80A, while the other three current levels were then determined based on the curve in Figure 6. In practice, the team began testing at 80A and then empirically found the other three levels of current failure.

The testing fixture, seen in Figure 21, itself was based around a 5V 300A DC power supply connected to a bank of 50 m Ω 50W discharge resistors to pull the desired current for a given test.

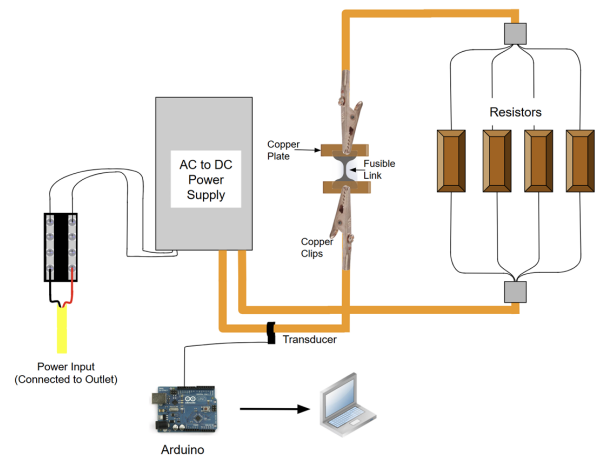


Figure 21: Fuse testing setup.

The current data was collected using a LEM DHAB hall effect current sensor connected to an Arduino microcontroller. The microcontroller sample period was set to 5 ms as per [4]. The serial time stamps were used to time each trial. A total of 6 tests were

performed over the first 3 current levels as per Figure 6.

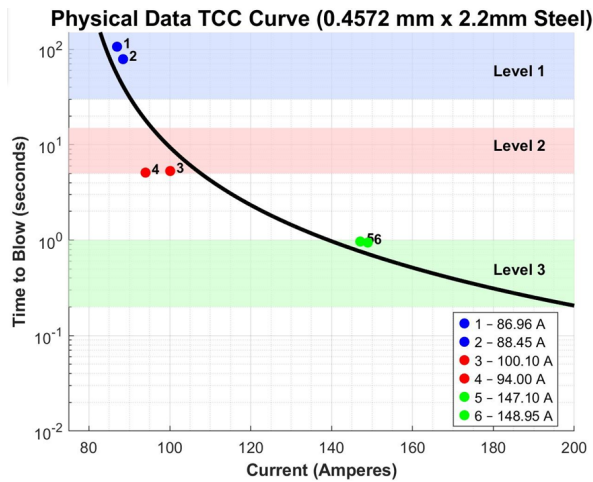


Figure 22: Fuse testing results data.

The team could not perform any tests at current level 4, as it is believed the internal resistance of the system was limiting the current draw from the power supply to around 160A. Approximate estimates for current level 4 requirements are around 200A. While testing, the team ensured safe practices due to extremely high currents such as insulating gloves and protective eyewear.

7.6 Discussion

In total, this team was successful in completing the entire design and testing process for a fusible link. Their work will act as a guide for future battery pack design for years to come. Some refinement in the testing process may be necessary due to the inability to reach high enough currents to test all failure ranges required by competition guidelines. Also, sourcing nickel plated steel for both testing and manufacturing purposes will be necessary for future seasons.

FSAE Rules Reference

T.7.1.1 Aerodynamic Device — A part on the vehicle which guides airflow for purposes including generation of downforce and/or change of drag.

EV.6.6.3 Each parallel element of multiple parallel battery cells, capacitors, strings of battery cells, strings of capacitors, or conductors must have individual Overcurrent Protection.

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